

Blood Smear AI Analysis

ResNet50 + Cellpose + Grad-CAM + SHAP + Vision-Language Explanation

This report presents the computational analysis of a microscopic blood-smear image using cell segmentation, five-class ResNet50 classification, and explainable-AI techniques.

Important: This is a research and screening-oriented computer-vision report. It is not a medical diagnosis.

1. Executive Summary

VLM explanation unavailable. The cell-level prediction and XAI visualizations are still included in this report.

The current analysis contains 6 selected cells for detailed XAI review.

2. Whole-Smear Analysis

The whole-smear stage provides the spatial context for the cell-level analysis. Cellpose segmentation identifies candidate objects, after which individual cell regions are passed to the ResNet50 classifier.

The whole-smear source image was not automatically located for embedding. The cell-level XAI evidence is still included.

3. Cell-Level Model Results

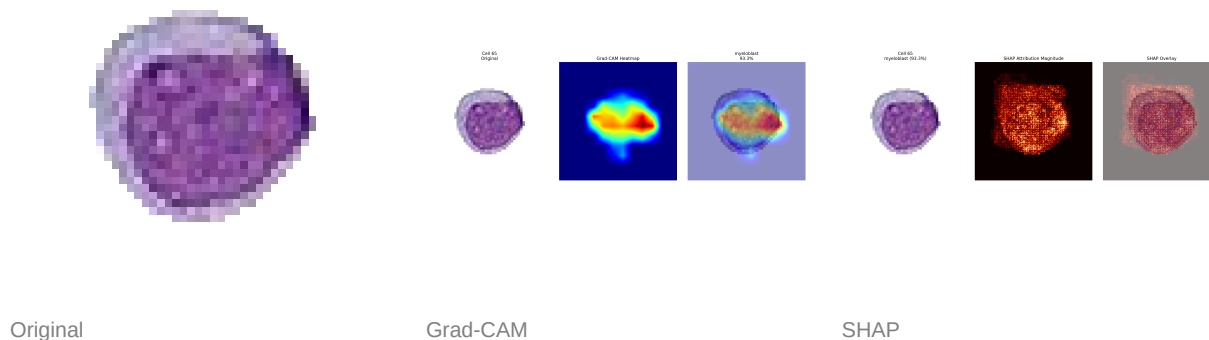
Cell	Predicted Class	Confidence	Grad-CAM	SHAP
65	myeloblast	0.9%	Available	Available
33	myeloblast	0.9%	Available	Available
4	myeloblast	0.9%	Available	Available
31	myeloblast	0.9%	Available	Available
52	myeloblast	0.9%	Available	Available
1	myeloblast	0.8%	Available	Available

The confidence values above are softmax classification scores. They should not be interpreted as clinical probabilities.

4. Explainable AI — Cell-by-Cell Analysis

Each selected cell is presented with the available original image, Grad-CAM visualization, and SHAP visualization. The vision-language model provides a constrained explanation of the computational evidence.

Cell 65 — myeloblast (0.9%)

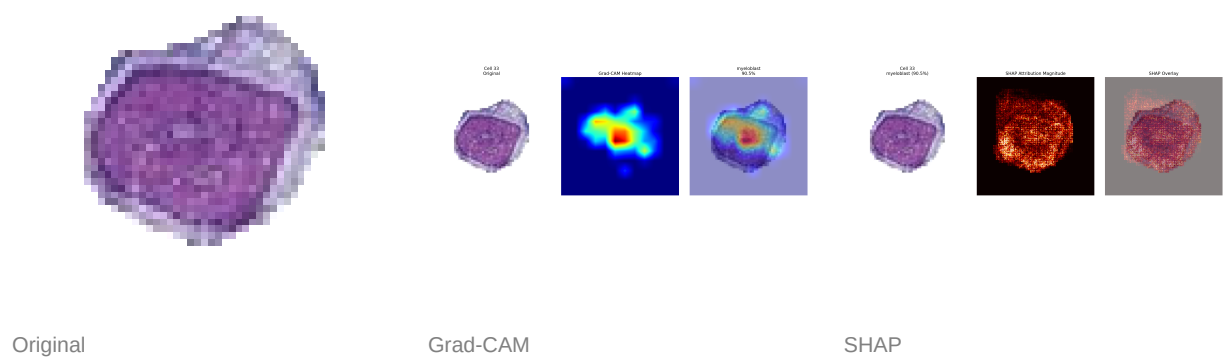


AI-generated explanation

VLM explanation unavailable. The cell-level prediction and XAI visualizations are still included in this report.

Interpretation note: XAI maps describe model attribution. They do not prove that a particular biological structure caused the prediction.

Cell 33 — myeloblast (0.9%)

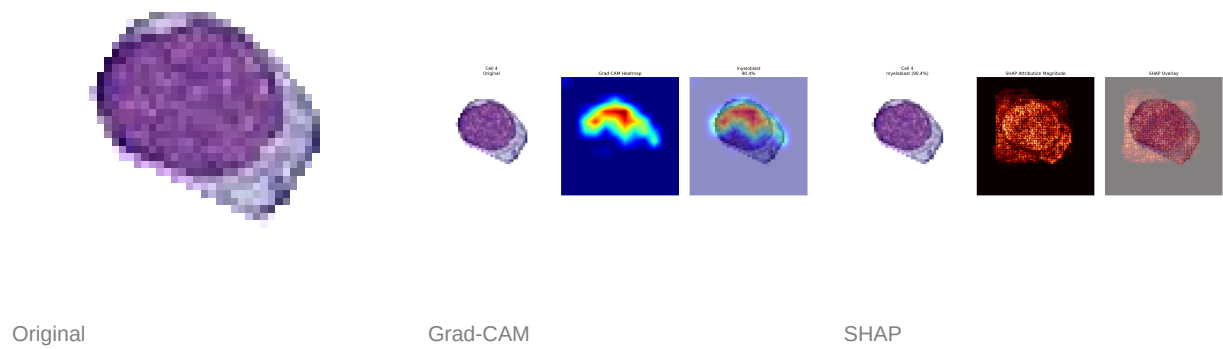


AI-generated explanation

VLM explanation unavailable. The cell-level prediction and XAI visualizations are still included in this report.

Interpretation note: XAI maps describe model attribution. They do not prove that a particular biological structure caused the prediction.

Cell 4 — myeloblast (0.9%)

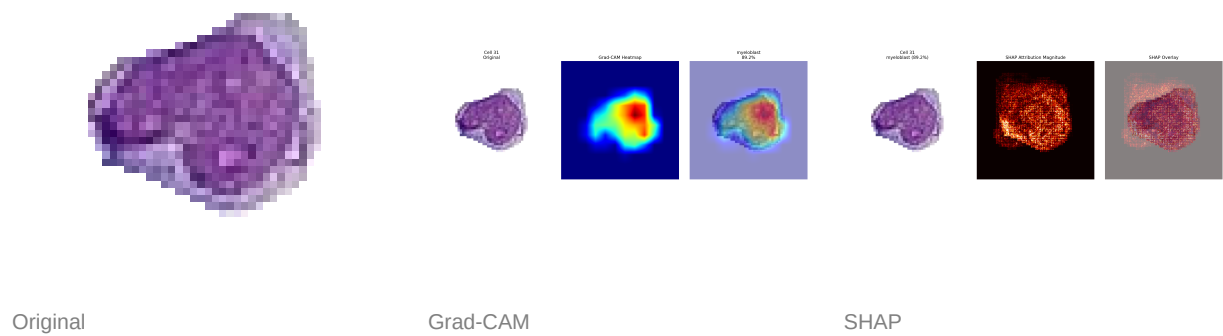


AI-generated explanation

VLM explanation unavailable. The cell-level prediction and XAI visualizations are still included in this report.

Interpretation note: XAI maps describe model attribution. They do not prove that a particular biological structure caused the prediction.

Cell 31 — myeloblast (0.9%)

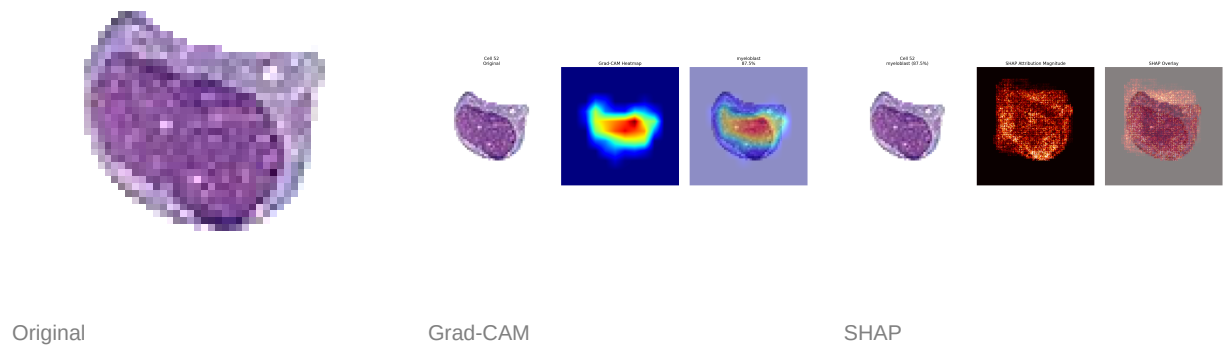


AI-generated explanation

VLM explanation unavailable. The cell-level prediction and XAI visualizations are still included in this report.

Interpretation note: XAI maps describe model attribution. They do not prove that a particular biological structure caused the prediction.

Cell 52 — myeloblast (0.9%)



AI-generated explanation

VLM explanation unavailable. The cell-level prediction and XAI visualizations are still included in this report.

Interpretation note: XAI maps describe model attribution. They do not prove that a particular biological structure caused the prediction.

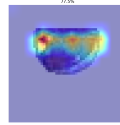
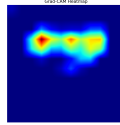
Cell 1 — myeloblast (0.8%)



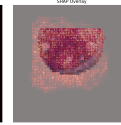
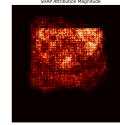
Original



Grad-CAM



SHAP



AI-generated explanation

VLM explanation unavailable. The cell-level prediction and XAI visualizations are still included in this report.

Interpretation note: XAI maps describe model attribution. They do not prove that a particular biological structure caused the prediction.

5. How the AI Evidence Should Be Interpreted

ResNet50

The classifier assigns each segmented cell to one of the five classes represented during training.

Cellpose

Cellpose provides the segmentation step that converts a whole smear into individual candidate cell regions.

Grad-CAM

Grad-CAM highlights spatial regions that contribute to the selected model output.

SHAP

SHAP provides pixel-level attribution for the selected class and can reveal whether model contribution is concentrated in cellular regions or extends into boundaries/background.

Vision-language model

Qwen2.5-VL is used here as a report-writing and evidence interpretation layer. It does not replace the trained ResNet50 classifier.

6. Limitations

- The ResNet50 classifier was trained on five cell categories rather than a cancer-versus-normal dataset.
- Cells outside the training categories can be forced into one of the available classes.
- A high softmax score does not guarantee biological correctness, particularly under domain shift.
- Grad-CAM and SHAP explain model behavior; they do not constitute proof of biological morphology.
- The current pipeline should not be used by itself to diagnose leukemia or another blood cancer.
- Clinical interpretation requires appropriate laboratory, morphological, immunophenotypic, molecular, and/or other diagnostic evidence as applicable.

7. Conclusion

The resulting system provides a multi-stage research pipeline that combines cell segmentation, deep-learning classification, and explainability. The addition of the vision-language layer turns the numerical predictions and XAI maps into a structured scientific explanation while preserving the distinction between machine-learning evidence and clinical diagnosis.

Research-use statement: This report is intended for research, model evaluation, and screening-oriented analysis. It is not a standalone medical diagnosis.